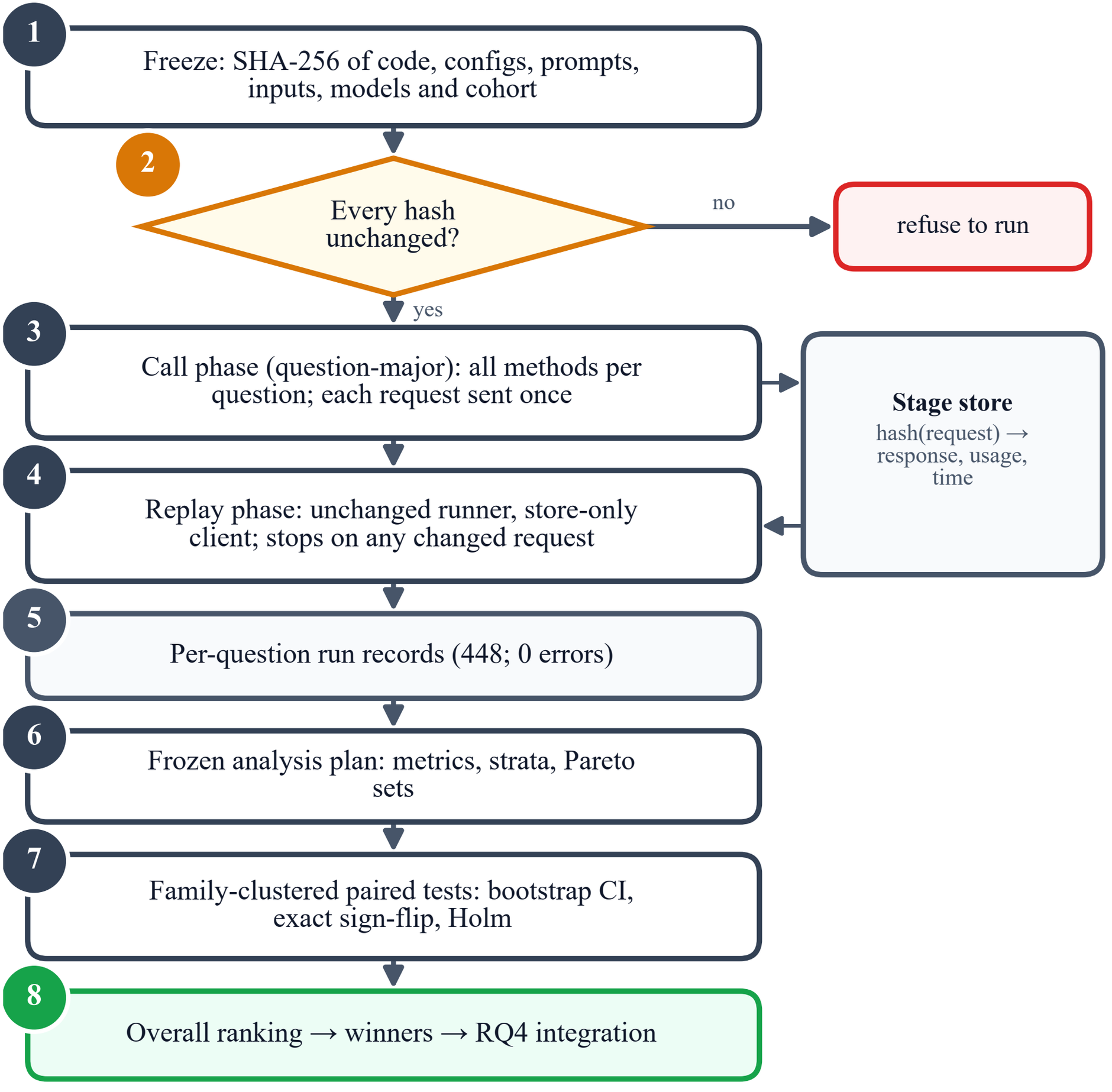

1
Freeze: SHA-256 of code, configs, prompts, inputs, models and cohort
2
Every hash unchanged?
no
refuse to run
yes
3
Call phase (question-major): all methods per question; each request sent once
4
Replay phase: unchanged runner, store-only client; stops on any changed request
Stage store
hash(request) → response, usage, time
5
Per-question run records (448; 0 errors)
6
Frozen analysis plan: metrics, strata, Pareto sets
7
Family-clustered paired tests: bootstrap CI, exact sign-flip, Holm
8
Overall ranking → winners → RQ4 integration